SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Research Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS COURSE CODE: MLA01

COURSE OUTCOME COVERED

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems.* (BTL 3)

Assessment Tool Weightage: 45 %

Total Marks: 50 **Time: 60 Mins**

No. of Questions: 5

Annexure A

Scenario-Based Research Assessment

Code of Conduct:

I Namburi Sai Yashaswini (192425250) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Yashaswini

Criterion	Weightage	Marks Obtained
Identification of Latest AI Application	10%	
Problem Analysis and Domain Understanding	15%	
AI Technologies and Working Mechanism	20%	
Research Quality and Literature Review	10%	
Real-World Implementation and Case Analysis	10%	
Impact, Challenges, and Future Scope	15%	
Originality and Critical Thinking	10%	
Organization, Presentation, and Documentation	10%	
Total	100%	

Signature of the Faculty Total Marks Awarded

ANSWER ANY ONE

No.	AI Domain	Scenario-Based Research Question
1	**Healthcare**	Healthcare organizations are increasingly adopting Artificial Intelligence to improve diagnosis, treatment, and patient monitoring. Identify and research a latest AI-based application in healthcare (such as AI-assisted medical diagnosis, AI radiology systems, robotic surgery, AI drug discovery, or virtual healthcare assistants). Analyze its working principles, AI techniques used, real-world implementation, benefits, and limitations.
2	**Finance**	Financial institutions are leveraging Artificial Intelligence to enhance security, decision-making, and customer services. Identify and research a latest AI application in the finance domain (such as AI fraud detection systems, AI-powered financial advisors, algorithmic trading platforms, or intelligent banking assistants). Discuss its functionality, technologies involved, advantages, and challenges.
3	**Retail**	Retail companies are adopting AI-driven solutions to provide personalized and intelligent shopping experiences. Identify and research a latest AI application in retail (such as Generative AI shopping assistants, AI recommendation engines, virtual try-on systems, or AI-based demand forecasting). Explain its application, AI methods used, impact on customers and businesses, and future scope.
4	**Autonomous Vehicles**	The transportation sector is rapidly evolving with AI-based autonomous systems. Identify and research a latest AI application in autonomous vehicles (such as self-driving technology, AI-based perception systems, autonomous delivery vehicles, or intelligent traffic management). Analyze the AI technologies involved, real-world deployment, benefits, and challenges.
5	**Robotics**	Intelligent robots are being developed for applications in industries, healthcare, education, and daily life. Identify and research a latest AI application in robotics (such as humanoid robots, collaborative robots, service robots, or AI-powered warehouse robots). Explain their capabilities, AI techniques, real-world usage and future developments.
6	**Manufacturing**	Manufacturing industries are integrating AI to achieve smart and automated production environments. Identify and research a latest AI application in smart manufacturing (such as AI-based predictive maintenance, computer vision quality inspection, digital twins, or autonomous manufacturing systems). Discuss its working, industrial applications, advantages, and limitations.

Humanoid Robots in Industry

AI-Driven Physical Intelligence for Manufacturing and Logistics

AI Domain Selected: Robotics (Question 5)

1. Title of AI Application

Humanoid Robots for Industrial and Logistics Automation — represented by platforms such as Tesla Optimus, Figure AI's Figure 03, Agility Robotics' Digit, Boston Dynamics' electric Atlas, and Apptronik's Apollo. The application belongs to the Robotics domain of Artificial Intelligence.

2. Domain Selection

Domain: Robotics. This report focuses specifically on general-purpose humanoid robots deployed in structured industrial environments — automotive assembly lines and logistics warehouses — where physical AI systems combine perception, planning, and learned motor control to perform tasks previously done only by human workers.

3. Introduction and Background

Humanoid robots are bipedal, human-shaped machines designed to operate in environments and use tools built for people, without requiring the world to be re-engineered around them. Although humanoid platforms have existed as research demonstrations for decades, 2025–2026 marks a turning point: several companies have moved from staged demo videos to sustained, paid, real-world deployments.

Tesla has reported over 50,000 cumulative Optimus units produced, with more than 1,000 units performing live production tasks at its Fremont facility. Figure AI's robots operated for roughly eleven months on BMW's Spartanburg line, contributing to over 30,000 vehicles, and expanded to BMW's Leipzig plant in March 2026. Agility Robotics' Digit has moved more than 100,000 totes at GXO warehouses and signed agreements with Toyota and Amazon.

4. Problem Identification

Warehouses and factories face chronic labour shortages, high injury rates from repetitive material-handling tasks, and rising wage costs, while most existing automation (fixed-arm robots, conveyor systems) only works in rigidly engineered environments. Reconfiguring a warehouse or line for wheeled or fixed robots is slow and expensive, and such systems cannot use existing human tools, ladders, or door handles.

The real-world problem is therefore: how can physically demanding, repetitive, or hazardous tasks in existing human-built spaces be automated without redesigning the entire facility around a robot's limitations.

5. Need for AI Solution

Traditional industrial robotics relies on fixed-path, pre-programmed motion within caged cells — effective for identical, repeated motions but brittle to variation. Humanoid robots need AI, not just mechanics, because they must perceive unstructured scenes, grasp objects of unknown shape, balance dynamically on two legs, and adapt instructions given in natural language.

- Traditional automation: rigid, pre-scripted, requires custom tooling per task.
- AI-driven humanoids: perceive, generalize, and learn new tasks from demonstration or language instruction.

- AI enables one physical platform to be repurposed across many tasks simply by updating its model, not its hardware.

6. AI Technologies Used

Modern humanoid robots integrate several AI subfields into one embodied system:

- Computer Vision — multi-camera perception for object recognition, depth estimation, and scene understanding (e.g., Tesla's vision-only, camera-based perception stack derived from its self-driving work).
- Deep Reinforcement Learning — trains locomotion and balance policies in simulation before transfer to real hardware.
- Imitation / Vision-Language-Action (VLA) Models — large neural networks (e.g., Figure's Helix system) that map camera input and language commands directly to motor actions, allowing robots to learn by watching humans instead of hand-coded rules.
- Natural Language Processing — lets operators issue task instructions in plain English rather than robot-specific code.
- Sensor Fusion and Control Theory — combines IMU, force, and torque sensor data for real-time balance and safe manipulation.

7. Working Mechanism

A humanoid robot's control loop can be summarized as: Input → Perception → Planning/Decision → Action → Feedback.

- Input: RGB cameras, depth sensors, joint encoders, and force-torque sensors continuously capture the robot's surroundings and its own body state.
- Processing: A perception model identifies objects and free space; a VLA or planning model decides the next motion primitive (e.g., 'pick up tote', 'walk to shelf B3').
- Decision-making: Learned policies, trained via reinforcement learning and human demonstration data, translate high-level goals into joint-level motor commands, adjusting in real time for balance.
- Output: Coordinated actuator commands drive the robot's limbs; outcomes (success, slip, drop) are logged and fed back to improve future performance.

8. Data Sources and Processing

Training data comes from three main sources: (1) teleoperated human demonstrations, where an operator remotely controls the robot to perform a task while sensor and video data are recorded; (2) large-scale video of humans performing tasks, used for imitation learning, echoing the same vision-only approach Tesla applies from its driver-camera fleet to Optimus; and (3) simulation, where physics engines generate millions of synthetic training episodes for locomotion and grasping before deployment on real hardware. This combined dataset is processed through neural networks that compress raw sensor streams into compact action decisions, and fleet data from deployed robots is continuously used to retrain and improve the shared model.

9. Real-World Implementation

Concrete, verified deployments as of mid-2026 include:

- Automotive: Figure AI robots on BMW's Spartanburg and Leipzig lines; Apptronik's Apollo tested at Mercedes-Benz's Berlin-Marienfelde plant; Tesla Optimus operating within Tesla's own Fremont and Gigafactory sites.
- Logistics and warehousing: Agility Robotics' Digit moving totes at GXO and Amazon warehouses, with agreements extending to Toyota; Figure AI robots in partner warehouse pilots.
- Enterprise-grade hardware: Boston Dynamics' all-electric Atlas, unveiled at CES 2026, allocated to Hyundai's Robotics Metaplant Application Center and Google DeepMind for advanced manipulation research.

Analysts at Goldman Sachs and Morgan Stanley project the humanoid robot market to reach roughly USD 38 billion by 2035 and USD 152 billion by 2040 respectively, reflecting expectations of continued industrial scaling.

10. Impact and Benefits

- Efficiency: robots can work continuous shifts without fatigue on repetitive material-handling tasks, increasing warehouse throughput.
- Safety: removes humans from high-injury-rate tasks such as repetitive heavy lifting.
- Flexibility: the same physical platform can be redeployed to new tasks via software updates rather than new hardware.
- Cost trajectory: vertically integrated manufacturers (e.g., Tesla) are targeting long-term unit costs comparable to a passenger vehicle, which could substantially lower automation cost-per-task over time.

11. Challenges and Limitations

- Technical: dynamic balance, dexterous grasping of irregular objects, and long-horizon task planning remain unsolved in unstructured settings.
- Economic: current commercial units (for example, Agility's Digit) cost far more than comparable fixed automation, and payback periods are still being proven.
- Safety and standards: industrial safety frameworks such as ISO 10218 must be adapted for robots that share open floor space with humans, rather than working in caged cells.
- Workforce impact: displacement of warehouse and assembly-line labour raises ethical and social questions requiring workforce transition planning.
- Data and privacy: continuous camera-based perception in workplaces raises surveillance and data-governance concerns.

12. Future Scope and Emerging Trends

- Falling hardware costs and higher-volume manufacturing are expected to push humanoid robot prices toward the tens-of-thousands-of-dollars range within a few years.
- Expansion from automotive and logistics into healthcare, retail, and eventually home settings as safety and dexterity improve.
- Growing use of foundation-model-based VLA systems to let one trained model generalize across many physical tasks and even robot bodies.
- Increased competition from Chinese manufacturers (Unitree, AgiBot, Fourier Intelligence) offering lower-cost hardware, intensifying global scaling of the technology.

13. Conclusion

Humanoid robots exemplify how combining computer vision, reinforcement learning, and large learned action models allows a single physical platform to generalize across tasks that once required custom automation for each use case. While still early — limited by cost, dexterity, and safety standards — 2026 marks the transition of humanoid robotics from research demonstration to measurable industrial deployment, with real production data from companies such as Tesla, Figure AI, and Agility Robotics. As AI models and manufacturing scale improve together, humanoid robots are positioned to become a significant contributor to solving labour shortages and hazardous-task automation across industries.

14. References

[1] AI Magicx Blog, "Humanoid Robots in the Workplace: The 2026 Business Leader's Reality Check," March 2026.

[2] EVST, "Top 8 Humanoid Robot Companies to Watch in 2026."

[3] VaaSBlock, "Humanoid Robotics 2026: Figure, Optimus, 1X Commercial Reality."

[4] Optimusk Blog, "What Is Tesla Optimus? Complete Guide to Tesla's Humanoid Robot (2025–2026)," updated July 2026.

[5] RoboZaps Blog, "38 Best Humanoid Robots in 2026 (Evidence-Ranked)."

[6] iFactory App, "Tesla Optimus at Fremont: Gen 3 Humanoid Deployment & Mass Production Update 2026."

[7] Humanoid Robotics Technology, "Top 12 Humanoid Robots of 2026."

[8] Technerdo, "Humanoid Robots in 2026: Market Leaders, Deployments, and What Comes Next," April 2026.

[9] New Market Pitch, "Figure 03 vs Tesla Optimus Comparison Tracker (2026

Rubrics for Evaluation

S.No	Criteria	Weightage (%)	Excellent (100–90%)	Good (89–75%)	Satisfactory (74–60%)	Improvement (
1.	Identification of Latest AI Application	10%	Selects a highly relevant, emerging AI application with strong justification.	Selects a recent and relevant AI application.	Selects a suitable application with limited justification.	Application is outdated or irrelevant.
2.	Problem Analysis and Domain Understanding	15%	Clearly explains the real-world problem, domain context, and need for AI with critical analysis.	Good explanation with adequate analysis.	Basic explanation with limited analysis.	Problem statement is unclear or incomplete.
3.	AI Technologies and Working Mechanism	20%	Accurately explains AI techniques, algorithms, architecture, and workflow with technical depth.	Good explanation with minor omissions.	Covers only basic AI concepts.	AI technologies are poorly explained or incorrect.
4.	Research Quality and Literature Review	10%	Uses recent, credible sources with proper citations and excellent synthesis.	Good quality references with appropriate citations.	Limited references with basic discussion.	Few or unreliable references; poor citation.
5.	Real-World Implementation and Case Analysis	10%	Thoroughly analyzes industrial implementation with appropriate examples.	Good implementation analysis.	Basic implementation discussion.	Minimal or no real-world analysis.
6.	Impact, Challenges, and Future Scope	15%	Critically evaluates benefits, limitations, ethical issues, and future trends.	Good evaluation with minor gaps.	Basic discussion of impact and future scope.	Limited evaluation or missing sections.
7.	Originality and Critical Thinking	10%	Demonstrates independent research, innovation, and critical evaluation.	Good analytical thinking.	Some originality with limited analysis.	Mostly descriptive or copied content.
8.	Organization, Presentation, and Documentation	10%	Exceptionally organized, professional presentation, error-free, and properly referenced.	Well organized with minor issues.	Adequately organized with some errors.	Poor organization and documentation.